

MATH 221 — Worksheet 3

1. calculus/derivatives/product-rule-01 *Product rule*

Not used before.

Let $h(x) = x^2e^x$.

(a) (3 points) Find $h'(x)$.

(b) (3 points) Find the equation of the tangent line at $x = 1$.

(c) (4 points) Find all critical points of h .

2. calculus/derivatives/chain-rule-01 *Chain rule with trig*

date	course	term	type	assignment
2026-10-12	MATH 221	Fall 2026	exam	Exam 1

Differentiate each function.

(a) (4 points) $f(x) = \sin(x^2)$

(b) (6 points) $g(x) = \cos^3(2x)$

3. (8 points) calculus/integrals/u-sub-01 *Substitution*

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Evaluate $\int x\sqrt{x^2+1} \, dx$.

4. linear-algebra/eigen-01 *Eigenvalues of a 2x2 matrix*

Not used before.

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

(a) Find the eigenvalues and eigenvectors of A .

i. (3 points) Eigenvalues.

ii. (3 points) Eigenvectors.

(b) (4 points) Diagonalize A .